

Seamoon Pandey

B.E. Computer Engineering | AI & Backend Engineer

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Pokhara, Nepal | Date of birth: 27 January 2003 | Nationality: Nepali

PROFILE

Computer Engineering graduate of the Institute of Engineering, Tribhuvan University, applying for an M.Sc. in Computer Science. My final-year thesis built a cross-site scripting scanner on a fine-tuned DistilBERT classifier and a microservice backend. I now work full-time as a full-stack engineer at Lamida Inc., building Web3 systems.

EDUCATION

Pashchimanchal Campus, Institute of Engineering, Tribhuvan University

Pokhara, Nepal

Bachelor of Engineering in Computer Engineering

2022 – 2026

Aggregate: **67.98%, First Division** | 1096.1 / 1612.5 marks | T.U. Regd. 3-2-45-173-2021

Tribhuvan University scale: Distinction $\geq 80\%$, First Division $\geq 65\%$, Second Division $\geq 50\%$. Examinations are set and moderated centrally by the university and marks are not scaled.

Thesis: RedSentinel, an AI-assisted cross-site scripting (XSS) vulnerability scanner. Lead architect, team of 3.

LANGUAGE PROFICIENCY

- **English:** IELTS Academic **7.5** overall, **CEFR C1** (15 August 2026). Listening 8.5 | Reading 7.5 | Writing 7.0 | Speaking 7.0. No band below 7.0.
- **Nepali:** native speaker. **Medium of instruction:** English, throughout the Bachelor's degree.

THESIS PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner

Final-year Thesis, Team of 3 | [Code](#) | [Report](#) | [Demo](#)

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Architected a polyglot microservice system: a **TypeScript/NestJS** core orchestrating three **Python/FastAPI** AI services over a **Redis/BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritise payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** Node.js, NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend & Security:** React, Next.js, Flutter; XSS detection, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

SELECTED PROJECTS

- **AI Artifact Recognition Platform** (PEC Hackathon: **YOLOv8**, Flask, React). Flask backend and React dashboard for real-time object recognition, with asynchronous inference and structured monitoring endpoints.
- **Community Connect** (OSM Hackathon: Express.js, Flutter). Full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.
- **Garva Diary** (BMC InnovateX: Ruby on Rails). Backend APIs on a Rails MVC architecture with structured validation workflows and mobile integration.

EXPERIENCE

Full-Stack Engineer, Lamida Inc.

Full-time | 23 Feb 2026 – Present

- Build full-stack Node.js applications for scalable Web3 products, including a DAO for the tUSDT token.
- **Team Lead, Bitfolio** from September 2026.

Freelance Full-Stack Developer

Contract, Remote

- Delivered production platforms on an **Express.js + React** stack (authentication, REST APIs, third-party integrations) while carrying a full course load: **Jyotisam**, an astrology consultation and booking platform (jyotisam.com), and **Mahadaan**, a donation and charity platform with secure data handling (mahadaan.com).

Mentor, Call for Enthusiasts Programme

Innovative Computer Engineering Students' Society (i-CES)

- Mentored junior students on scalable application design, clean backend architecture, and development practice.